

# Sukhman Balagan

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## Education

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**York University**

*Bachelor of Science in Computer Science*

**Toronto, ON**

**September 2024 – Present**

## Technical Skills

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**Programming Languages:** Python, Java, HTML, CSS, JavaScript, C#, C, Ruby

**Software & Tools:** GitHub, Azure DevOps, AWS, PostgreSQL, REST APIs, OAuth, PowerBI, CI/CD

**Frameworks & Libraries:** React, Next.js, Node.js, Express, Vite, TailwindCSS, TensorFlow, SQL

## Experience

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**PTAG**

**Markham, ON**

*Software Engineer Intern*

**May 2026 – August 2026**

- Designed and shipped Outfitter, a full-stack internal platform to discover and govern firm-wide AI tool usage, owning it end-to-end from data model to production deployment.
- Built a React 18 + TypeScript frontend and a Node.js + Express REST API over PostgreSQL with Microsoft Entra (MSAL) OAuth, a searchable catalog, and a Trelica-fed analytics dashboard.
- Deployed to Azure App Service via a GitHub Actions CI/CD pipeline with Key Vault secrets, managed identity, and PostgreSQL Flexible Server.
- Delivered the platform as its sole developer ~3–4 weeks ahead of schedule, and authored a full handoff document covering the data model, API, frontend, and Azure infrastructure.

*Software Engineer Intern*

**May 2025 – August 2025**

- Built a full-stack embedded analytics feature integrating Power BI into the PTAG FM web app using Azure AD, Power BI Embedded, and the JavaScript SDK.
- Developed Ruby on Rails REST controllers and backend services for secure embedding with RBAC.
- Built Power BI data pipelines with Power Query and DAX to turn JSON APIs into real-time KPI dashboards, shipped via Azure DevOps CI/CD.

**The Groundwater Project**

**Remote**

*Full-Stack Developer (Contract)*

**May 2026 – June 2026**

- Built AI-powered semantic search over the organization's educational library, letting readers ask natural-language questions across its materials.
- Engineered the RAG retrieval pipeline: Supabase/pgvector for embeddings, MeiliSearch for keyword and hybrid retrieval, and Meta NLLB for multilingual support, including African languages.
- Built the search UI and documented the algorithm, architecture, and user-testing/evaluation results.

## Projects

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**DiamondAI** | [Website](#) | [Code](#) | *AWS (SageMaker, Lambda, Kinesis, DynamoDB, API Gateway), CDK, JAX* **June 2026**

- Designed an event-driven serving architecture on AWS: Lambda streams live MLB events into Kinesis, a SageMaker endpoint serves model inference with autoscaling, and DynamoDB Streams drive an automated settlement pipeline.
- Pushed sub-second predictions to the frontend over an API Gateway WebSocket API; defined all infrastructure as code in AWS CDK (TypeScript) with CloudWatch and X-Ray observability.
- Trained the underlying ~7.5M-param Transformer from scratch on ~7M pitches on a Kaggle TPU.

**SideCode** | [Website](#) | *Gemini API, Vertex AI, Firebase, GitHub API, Monaco*

**June 2026**

- Built a multi-agent code-review companion that renders a GitHub repo as an interactive graph and streams per-file security and bug verdicts in real time via Firestore listeners, with one-click fixes.
- Ran parallel Gemini agents with a coordinator reconciling findings into typed JSON; a webhook rescans changed files on every push.
- Runner-up at the GDG York AI Case Competition (Google Toronto HQ).

**HomeCrowd** | *NVIDIA GB10, Nemotron 30B, cuOpt, FastAPI, Deck.gl*

**May 2026**

- Built a local-first AI system on NVIDIA GB10 Grace Blackwell that forecasts neighborhood event impact for Toronto small businesses and generates day-by-day staffing and inventory plans.
- Ran Nemotron 30B locally with cuOpt GPU optimization behind a FastAPI + Deck.gl dashboard.
- Named a finalist at the NVIDIA Spark Hackathon.